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Effect of workplace incivility on nurses' health outcomes: a cross-sectional study

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Abstract

Background Workplace incivility is increasingly recognized as a key occupational stressor in nursing. However, few studies have explored its independent association with nurse health after adjustment for established psychosocial work characteristics. This study examined the associations of workplace incivility from colleagues and supervisors with nurses' self-rated health, burnout, and mental health status after adjusting for job control, psychological job demands, and workplace justice.

Methods This cross-sectional study was conducted between October and December 2022 at a medical center in northern Taiwan. Full-time registered nurses completed a self-administered questionnaire. A total of 900 valid responses were analyzed (response rate, 84.0%). The questionnaire assessed workplace incivility from colleagues and supervisors, personal burnout, mental health status, self-rated health, and psychosocial work characteristics (job control, psychological job demands, and workplace justice). The chi-square test and multivariable logistic regression were used to examine potential associations between workplace incivility and health outcomes after adjustment for demographic and psychosocial work characteristics.

Results The prevalence of poor mental health, high personal burnout, and poor self-rated health was higher in nurses with high incivility exposure than in those with low incivility exposure (51.8% vs. 30.0%, 55.3% vs. 40.1%, and 20.4% vs. 11.9%, respectively). After adjustment for psychosocial work characteristics and demographic factors, high workplace incivility remained significantly associated with poor mental health, high personal burnout, and poor self-rated health. High psychological job demands were also significantly associated with increased risks of personal burnout, poor mental health, and poor self-rated health.

Conclusions Workplace incivility appears to be significantly associated with adverse health outcomes in nurses, independent of psychosocial work characteristics and demographic factors. Furthermore, high psychological job demands may be significantly associated with personal burnout, poor mental health, and poor self-rated health in nurses.

Impact The findings highlight workplace incivility as a key occupational health problem in nursing. Strategies to address workplace incivility and psychological job demands and promote a healthier work environment may support nurses' mental health and overall well-being. This study may guide nurse managers and policymakers in developing organizational strategies to support workplace health and safety.

Keywords Workplace incivility · Self-rated health · Mental health · Burnout · Nursing

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Introduction

Workplace psychosocial hazards are major occupational health concerns in health-care workers, particularly nurses [1]. Interpersonal conflicts and uncivil workplace interactions are key psychosocial stressors associated with adverse health and well-being [2]. A meta-analysis reported that approximately 34% of all employees experienced workplace mistreatment and 44% witnessed such behaviors. Prevalence estimates for different forms of workplace mistreatment ranged from 16% to 75% depending on the study population and measurement approach [3]. Among various forms of workplace mistreatment, workplace incivility is particularly common in health-care settings [4]. Workplace incivility is defined as low-intensity deviant behavior with an ambiguous intent to harm that violates workplace norms of mutual respect [5]. It differs from bullying and workplace violence in its relatively low intensity and ambiguous intent [6, 7]. In health-care settings, workplace incivility may come from colleagues, supervisors, interdisciplinary team members, and patients or their family members and typically involves rude and discourteous behavior that exhibits a lack of regard for others [8]. Nurses comprise approximately 59% of the health-care workforce and are particularly vulnerable to workplace incivility because their work involves frequent interpersonal interactions [9]. A systematic review and meta-analysis reported that the global prevalence of incivility toward nurses was 55.1% and that verbal incivility was the most common form [10]. Studies conducted in different settings have reported prevalence estimates ranging from 67.5% to 90.4% [11]. In Taiwan, 41.1% of all nurses face high workplace incivility [12], and nurses experience an average of 1.27 incidents of workplace incivility per workday [13]. Repeated exposure to uncivil behavior may normalize disrespectful interactions, reduce interpersonal trust, and increase the risk of escalation to more severe forms of workplace mistreatment, such as bullying and violence. Therefore, early recognition and effective management of workplace incivility are crucial for preventing such escalation and maintaining a respectful work environment [4, 14, 15]. The present study specifically focused on incivility from colleagues and supervisors because these interactions occur within ongoing professional and organizational relationships and may cause more sustained psychological strain than do occasional interactions with patients or their family members.

Several studies have associated workplace incivility with adverse health outcomes, suggesting that workplace incivility acts as a chronic interpersonal stressor with both immediate and cumulative effects on employee well-being [7, 8, 16]. In nurses, workplace incivility has been associated with poor self-rated health [17], fatigue [18], sleep disturbances [19], poor mental health [20], and increased turnover intentions [21]. In addition, workplace incivility has been associated with psychological stress, emotional exhaustion, and physiological stress responses, which may adversely affect nurses' health and work performance. Beyond its effects on individual nurses, workplace incivility may disrupt communication, teamwork, and work engagement, with potential consequences for patient safety and care quality. Workplace incivility has also been associated with increased 30-day readmission rates and reduced patient satisfaction [8]. Prolonged exposure to workplace incivility has been linked to poorer physical and mental health in nurses [22, 23]. Despite these findings, several gaps remain in research on social incivility. Some studies have combined workplace incivility with other negative workplace behaviors or distinguished it from bullying primarily on the basis of frequency [24, 25]. Other studies have classified abuse, threats, and insults as incivility [26–28]. Furthermore, many studies have not adequately accounted for psychosocial work characteristics, such as psychological job demands, job control, and workplace justice, which are established determinants of nurse health [29]. According to the Job Demands–Resources model, employee well-being is affected by the balance between job demands and job resources. Psychological job demands are aspects of work that require sustained psychological or emotional effort. By contrast, job control and workplace justice are key psychosocial resources that may help employees cope with workplace stressors. Because these psychosocial work characteristics are well-established determinants of employee health, they may confound the association between workplace incivility and health outcomes [30]. Failure to account for these factors may therefore result in biased estimates of the association between workplace incivility and health outcomes [8]. Accordingly,

the present study examined the associations of workplace incivility with nurses' self-rated health, burnout, and mental health while accounting for psychological job demands, job control, and workplace justice.

The study

Aim

This study examined the associations of workplace incivility with nurses' self-rated health, burnout, and mental health.

Research questions

- Is workplace incivility associated with nurses' self-rated health, burnout, and mental health?
- If yes, do the associations persist after adjustment for psychological job demands, job control, workplace justice, and demographic factors?
- Which demographic and work characteristics are associated with increased exposure to workplace incivility among nurses?

Methods

Design

This cross-sectional study was conducted between October and December 2022 at a medical center in northern Taiwan.

Participants

A total of 1067 questionnaires were distributed across 38 nursing units. A total of 900 valid responses were obtained, yielding a response rate of 84.0%. Nurses were eligible if they were employed full-time in clinical care and could complete the questionnaire in Chinese. Exclusion criteria were part-time employment, employment in administrative or nonclinical roles, extended leave, and questionnaires with substantial missing data.

Data collection

Workplace incivility

Workplace incivility is a form of interpersonal deviance that violates workplace ethics or social interaction norms [5]. Despite its relatively mild nature, workplace incivility can have long-term adverse effects on team dynamics and individual well-being. In the present study, workplace incivility was measured using the 7-item Workplace Incivility Scale developed by Cortina et al. [31]. Participants were asked about uncivil behaviors they had experienced during the previous 12 months in their current work unit. The scale specifically assessed incivility from supervisors or colleagues and excluded incivility from patients or their family members. The seven items assessed whether, during this period, a supervisor or colleague had (1) demeaned that participant or acted condescendingly toward them, (2) paid little attention to their statements or displayed little interest in their opinion, (3) made demeaning or insulting remarks about them, (4) addressed them using inappropriate terms, either publicly

or privately, (5) ignored or excluded them from professional camaraderie, (6) doubted their judgment on matters within their area of responsibility, or (7) involved them in discussions of personal matters against their will [12]. The Chinese version of the Workplace Incivility Scale was developed through forward and backward translation and demonstrated good reliability and structural validity in a validation sample of 355 full-time ward nurses [12]. In the current sample, the Chinese version exhibited excellent internal consistency (Cronbach's $\alpha = 0.951$). Items were rated on a 5-point Likert scale with endpoints ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). Higher scores indicated greater exposure to workplace incivility [12]. For subsequent analyses, workplace incivility scores were dichotomized into low- and high-incivility groups by using a median split, consistent with the approach used in a prior study involving Taiwanese nurses [12].

Personal burnout

Burnout was measured using the personal burnout subscale of the Copenhagen Burnout Inventory [32], which has been validated for use in Chinese populations [33]. The subscale consists of five items that assess fatigue, physical exhaustion, and emotional exhaustion. Participants were asked to report their experiences during the previous week. Each item was rated on a 5-point scale with endpoints ranging from 0 (*never*) to 4 (*always*). The item scores were summed and multiplied by 5 to obtain a total score ranging from 0 to 100. Higher scores indicated greater personal burnout. Previous validation studies in Taiwanese health-care workers have demonstrated good internal consistency, with Cronbach's α values ranging from 0.86 to 0.92 [33–35]. In the present study, Cronbach's α was 0.92, indicating high internal consistency. On the basis of established cutoff criteria used in studies involving Taiwanese health-care workers, a score of ≥ 51 was classified as high burnout [35, 36], whereas a score of < 51 was classified as low burnout.

Mental health status

Mental health status refers to emotional and psychological experiences, including anxiety, depression, stress, and other emotional difficulties that can affect well-being and functioning [37, 38]. In the present study, mental health was assessed using the 5-item Brief Symptom Rating Scale developed by Lee et al. [37]. This scale was derived from the original 50-item symptom checklist. Participants were asked to report symptoms experienced during the previous week. Example items include "Did you feel tense or anxious?" and "Did you feel depressed or low in mood?" The remaining items assess irritability or anger, feelings of inferiority, and trouble sleeping. Each item was rated on a 5-point Likert scale with endpoints ranging from 0 (*not at all*) to 4 (*extremely*). Previous studies have demonstrated good reliability and validity of the 5-item Brief Symptom Rating Scale, with Cronbach's α values ranging from 0.77 to 0.90 and a test–retest reliability of 0.82 [37, 38]. In the present study, Cronbach's α was 0.90. Consistent with previous studies, a total score of ≥ 6 was classified as mental distress [35, 39, 40], whereas a score of < 6 was classified as good mental health.

Self-rated health

Self-rated health is a subjective measure of an individual's overall health status and is widely used in population and occupational health research [41, 42]. It is typically assessed using a single question, which has demonstrated good predictive validity for subsequent health outcomes [43, 44]. Participants were asked, "In general, how would you rate your current health status?" Responses were recorded on a 5-point scale with endpoints ranging from 1 (*very good*) to 5 (*very poor*). Higher scores indicated poorer health. A score of ≤ 3 was classified as good self-rated health, whereas a score of > 3 was classified as poor self-rated health [39, 40]. Previous studies conducted in Taiwanese employees and nurses have used the same cutoff to define poor self-rated health [40, 45].

Work characteristics

Psychological job demands were measured using the Job Content Questionnaire developed by Hurrell et al. [46]. This questionnaire comprises seven items that assess work pace, workload, and time pressure. Each item is rated on a 4-point Likert scale with endpoints ranging from 1 (*strongly disagree*) to 4 (*strongly agree*). Raw scores are calculated by summing the item scores after reverse coding negatively worded items. The scores are then standardized using the following formula: [(raw score − 7)/(28 − 7)] × 100%. Higher scores indicate greater psychological job demands. Previous studies have demonstrated good reliability and validity [34, 47, 48], with Cronbach's α values ranging from 0.78 to 0.8. In the present study, Cronbach's α was 0.80. Scores were categorized as low, medium, or high on the basis of tertiles.

Job control was measured using the Chinese version of the Job Content Questionnaire developed by Cheng et al. [49]. This questionnaire comprises nine items covering two dimensions: skill discretion and decision authority. Each item is rated on a 4-point Likert scale with endpoints ranging from 1 (*strongly disagree*) to 4 (*strongly agree*). The skill discretion score is calculated by summing the six skill discretion items and multiplying the result by 2, whereas the decision authority score is calculated by summing the three decision authority items and multiplying the result by 4. The total job control score is calculated as the sum of the skill discretion and decision authority scores. The total score is then standardized using the following formula: [(raw score − 24)/(96 − 24)] × 100%. Higher scores indicate greater job control. Previous studies have reported Cronbach's α values of 0.71 and 0.69 for skill discretion and decision authority, respectively, and 0.8 for the overall scale [40, 47, 49]. In the present study, Cronbach's α was 0.62. Scores were categorized as low, medium, or high on the basis of tertiles.

Workplace justice was measured using a scale developed by Colquitt [50] and Moorman [51]. The Chinese version of this scale was developed by Cheng et al. [34]. Items are rated on a 4-point Likert scale with endpoints ranging from 1 (*strongly disagree*) to 4 (*strongly agree*). The scale assesses employees' perceptions of fairness in organizational procedures, the distribution of outcomes, and interpersonal treatment. Previous studies have demonstrated good reliability and validity [40, 47, 49], with a reported Cronbach's α value of 0.87. In the present study, Cronbach's α was 0.92. Scores were standardized and categorized as low, medium, or high on the basis of tertiles.

Demographic characteristics

Data on participants' self-reported demographic characteristics were collected. These data included sex, age, job seniority, marital status, education level, and work shift.

Data analysis

All statistical analyses were conducted using IBM SPSS Statistics version 25.0 (IBM, Armonk, NY, USA). Descriptive statistics were used to summarize participants' demographic characteristics, work characteristics, and health outcomes. All variables are presented as frequencies and percentages.

The chi-square test was used to examine the associations between categorical variables and workplace incivility level (low vs. high). To identify factors associated with workplace incivility, multivariable logistic regression was performed to examine the associations of demographic and work characteristics with workplace incivility.

To examine the associations between workplace incivility and health outcomes, three separate multivariable logistic regression models were constructed, with high burnout, poor mental health, and poor self-rated health as the dependent variables [52, 53]. In each model, workplace incivility was the primary independent variable. Demographic characteristics (sex, education level, marital status, job seniority, and work shift) and work characteristics (job control, psychological job demands, and workplace justice) were included simultaneously as covariates. These covariates were selected on the basis of previous studies indicating their associations with health outcomes in nurses [34–36, 39]. Results are reported as odds ratios (ORs) with 95% confidence intervals

(CIs). A p value of < 0.05 was considered statistically significant. Hierarchical logistic regression models were constructed to examine the incremental variance explained by each set of variables. Variables were entered cumulatively: demographic characteristics in Model 1, psychosocial work characteristics in Model 2, and workplace incivility in Model 3 (Additional File 1).

Results

Table 1 presents the demographic characteristics of 900 participants stratified by workplace incivility level (low vs. high). No significant between-group difference was observed in sex ($p=0.271$), and most participants were women. Furthermore, no significant between-group difference was observed in job seniority (years, $p=0.382$).

Table 1 Distributions of the demographic characteristics, work characteristics, and health outcomes in the study cohort ($N=900$)

	Low Workplace Incivility ($n=464$)	High Workplace Incivility ($n=436$)	P value	Total
	N (%)	N (%)		N (%)
Sex			0.271	
Male	39 (8.4%)	46 (10.6%)		85 (9.4%)
Female	425 (91.6%)	390 (89.4%)		815 (90.6%)
Job seniority (year)			0.382	
≤1	110 (23.7%)	103 (23.6%)		213 (23.7%)
2–5	148 (31.9%)	122 (28.0%)		270 (30.0%)
>5	206 (44.4%)	211 (48.4%)		417 (46.3%)
Marital status			0.474	
Single	88 (19.0%)	91 (20.9%)		179 (19.9%)
Married	376 (81.0%)	345 (79.1%)		721 (80.1%)
Education			0.141	
Junior college or Vocational high school	143 (30.8%)	115 (26.4%)		258 (28.7%)
University and above	321 (69.2%)	321 (73.6%)		642 (71.3%)
Work shift			0.782	
Day shift	82 (17.7%)	74 (17.0%)		156 (17.3%)
Rotated/Night shift	382 (82.3%)	362 (83.0%)		744 (82.7%)
Job control			<0.001	
Low	114 (24.6%)	161 (36.9%)		275 (30.6%)
Medium	144 (31.0%)	161 (36.9%)		305 (33.9%)
High	206 (44.4%)	114 (26.1%)		320 (35.6%)
Psychological job demand			0.012	
Low	191 (41.2%)	139 (31.9%)		330 (36.7%)
Medium	138 (29.7%)	141 (32.3%)		279 (31.0%)
High	135 (29.1%)	156 (35.8%)		291 (32.3%)
Workplace justice			<0.001	
Low	101 (21.8%)	208 (47.7%)		309 (34.3%)
Medium	256 (55.2%)	178 (40.8%)		434 (48.2%)
High	107 (23.1%)	50 (11.5%)		157 (17.4%)
Burnout			<0.001	
Low	278 (59.9%)	195 (44.7%)		473 (52.6%)
High	186 (40.1%)	241 (55.3%)		427 (47.4%)
Mental health			<0.001	
Good	325 (70.0%)	210 (48.2%)		535 (59.4%)
Bad	139 (30.0%)	226 (51.8%)		365 (40.6%)
Self-rated health			<0.001	
Good	409 (88.1%)	347 (79.6%)		756 (84.0%)
Bad	55 (11.9%)	89 (20.4%)		144 (16.0%)

The proportions of participants with ≤1 year, 2–5 years, and >5 years of job seniority in the low- and high-incivility groups were 23.7% vs. 23.6%, 31.9% vs. 28.0%, and 44.4% vs. 48.4%, respectively. No between-group difference was observed in marital status or education level ($p=0.474$ and $p=0.141$, respectively). The distribution of work shifts was similar between the two groups ($p=0.782$), with most participants working rotating or night shifts.

The low-incivility group reported significantly higher job control ($p<0.001$), whereas the high-incivility group reported significantly higher psychological job demands ($p=0.012$) and lower workplace justice ($p<0.001$). Furthermore, the low-incivility group was more likely than the high-incivility group to report good self-rated health, low burnout, and good mental health (all $p<0.001$).

Table 2 lists factors associated with workplace incivility among nurses. After adjustment for demographic and work characteristics, job control and workplace justice were significantly associated with workplace incivility. Nurses with medium job control were more likely than those with high job control to experience workplace incivility (OR=2.035, 95% CI: 1.446–2.865, $p<0.001$). Similarly, nurses with low job control were more likely than those with high job control to experience workplace incivility (OR=2.096, 95% CI: 1.471–2.985, $p<0.001$).

Workplace justice was strongly associated with workplace incivility. Nurses reporting low workplace justice were nearly four times more likely than those reporting high workplace justice to experience workplace incivility (OR=3.874, 95% CI: 2.514–5.969, $p<0.001$). Medium workplace justice was not significantly associated with workplace incivility (OR=1.423, 95% CI: 0.948–2.136). No significant associations were observed between workplace incivility and demographic characteristics such as sex, education level, marital status, job seniority, and work shift. Similarly, no significant associations were noted between workplace incivility and psychological job demands.

Table 3 presents the results of the multivariable logistic regression analyses examining the associations between workplace incivility and three health outcomes: high personal burnout, poor mental health, and poor self-rated health. The Nagelkerke R^2 values were 0.247 for high personal burnout, 0.136 for poor mental health, and 0.128 for poor self-rated health.

For high personal burnout, nurses working rotating or night shifts were significantly less likely than those working day shifts to report high burnout (OR=0.546, 95% CI: 0.362–0.823). High workplace incivility was

Table 2 Adjusted odds ratios for demographic and work characteristics associated with workplace incivility in nurses ($N=900$)

	High Workplace Incivility		
	odds ratio	(95% CI)	
Sex (ref. male)			
Female	0.774	(0.477	1.257)
Education (ref. Junior college or Vocational high school)			
University and above	1.215	(0.891	1.656)
Marital status (ref. single)			
Married	1.028	(0.693	1.525)
Job seniority (year, ref. 5 and above)			
≤1	1.087	(0.733	1.612)
2–5	0.765	(0.538	1.088)
Work shift (ref. day shift)			
Rotated/Night shift	0.871	(0.592	1.280)
Job control (ref. high)			
Medium	2.035***	(1.446	2.865)
Low	2.096***	(1.471	2.985)
Psychological job demand (ref. low)			
Medium	1.304	(0.922	1.843)
High	1.327	(0.934	1.885)
Workplace justice (ref. high)			
Medium	1.423	(0.948	2.136)
Low	3.874***	(2.514	5.969)

Significance: $^{*}p<0.05$, $^{**}p<0.01$, and $^{***}p<0.001$. Abbreviations: ref., reference group; CI, confidence interval

Table 3 Adjusted odds ratios for demographic and work characteristics associated with high burnout, poor mental health, and poor self-rated health in nurses (N=900)

	High burnout			Poor mental health			Poor self-rated health		
	odds ratio	(95% CI)		odds ratio	(95% CI)		odds ratio	(95% CI)	
Sex (ref. male)									
Female	0.969	(0.585,	1.606)	1.190	(0.724,	1.954)	0.838	(0.443,	1.584)
Education (ref. Junior college or Vocational high school)									
University and above	0.725	(0.524,	1.003)	0.917	(0.670,	1.255)	0.868	(0.577,	1.307)
Marital status (ref. single)									
Married	1.481	(0.983,	2.232)	1.002	(0.672,	1.494)	0.879	(0.514,	1.502)
Job seniority (year, ref. 5 and above)									
≤1	1.174	(0.773,	1.783)	1.626*	(1.091,	2.421)	1.059	(0.616,	1.821)
2–5	1.020	(0.705,	1.475)	1.035	(0.723,	1.482)	1.383	(0.873,	2.192)
Work shift (ref. day shift)									
Rotated/Night shift	0.546**	(0.362,	0.823)	0.699	(0.474,	1.030)	0.735	(0.437,	1.236)
Job control (ref. high)									
Medium	1.134	(0.789,	1.629)	1.225	(0.863,	1.739)	0.806	(0.492,	1.321)
Low	1.150	(0.791,	1.672)	0.925	(0.642,	1.331)	1.184	(0.748,	1.874)
Psychological job demand (ref. low)									
Medium	3.205***	(2.228,	4.611)	1.622**	(1.133,	2.322)	1.848*	(1.066,	3.203)
High	7.573***	(5.170,	11.093)	2.833***	(1.977,	4.061)	3.994***	(2.395,	6.658)
Workplace justice (ref. high)									
Medium	0.999	(0.659,	1.514)	0.821	(0.547,	1.232)	0.893	(0.504,	1.582)
Low	1.544	(0.984,	2.425)	1.063	(0.688,	1.644)	1.483	(0.840,	2.619)
Workplace Incivility (ref. low)									
High	1.546**	(1.137,	2.102)	2.364***	(1.756,	3.182)	1.589*	(1.065,	2.369)
Nagelkerke R^2	0.247			0.136			0.128		

Significance: $^{*}p<0.05$, $^{**}p<0.01$, and $^{***}p<0.001$. Abbreviations: ref., reference group; CI, confidence interval

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significantly associated with increased odds of high personal burnout (OR = 1.546, 95% CI: 1.137–2.102). Nurses with medium psychological job demands had significantly greater odds of high burnout than did those with low psychological job demands (OR = 3.205, 95% CI: 2.228–4.611). Similarly, nurses with high psychological job demands had significantly greater odds of high burnout than did those with low psychological job demands (OR = 7.573, 95% CI: 5.170–11.093). High personal burnout was not significantly associated with sex, education level, marital status, job seniority, job control, or workplace justice.

For poor mental health, nurses with ≤1 year of experience had significantly greater odds of reporting poor mental health than did those with >5 years of experience (OR = 1.626, 95% CI: 1.091–2.421). Work shift was not significantly associated with poor mental health (OR = 0.699, 95% CI: 0.474–1.030). Nurses exposed to high workplace incivility had significantly greater odds of poor mental health than did those exposed to low workplace incivility (OR = 2.364, 95% CI: 1.756–3.182). Nurses with medium and high psychological job demands had significantly greater odds of poor mental health than did those with low psychological job demands (OR = 1.622 [95% CI: 1.133–2.322] and 2.833 [95% CI: 1.977–4.061], respectively). Poor mental health was not significantly associated with sex, education level, marital status, job control, or workplace justice.

Nurses exposed to high workplace incivility had significantly greater odds of poor self-rated health than did those exposed to low workplace incivility (OR = 1.589, 95% CI: 1.065–2.369). Furthermore, nurses with medium and high psychological job demands had significantly greater odds of poor self-rated health than did those with low psychological job demands (OR = 1.848 [95% CI: 1.066–3.203] and 3.994 [95% CI: 2.395–6.658], respectively). Poor self-rated health was not significantly associated with sex, education level, marital status, job seniority, work shift, or workplace justice.

Discussion

The present study examined the association between workplace incivility and nurses' self-rated health, burnout, and mental health. Nurses exposed to high workplace incivility were more likely than those exposed to low workplace incivility to report high burnout, poor mental health, and poor self-rated health. These associations remained significant after adjustment for psychological job demands, job control, and workplace justice, indicating that workplace incivility was associated with health outcomes in nurses independently of psychosocial work characteristics. In the descriptive analysis, 48.4% of all nurses were classified into the high-incivility group. This proportion was higher than the that reported by Bao et al. (41.1%) [12] but lower than that reported by Bambi et al. (67.5%–90.4%) [11]. These differences may reflect variations in study populations, clinical settings, and measurement approaches. Research suggests that nurses perceive workplace incivility as a source of distraction and emotional distress, with potential implications for patient safety. Lee et al. [29] reported that workplace incivility and workload were associated with increased burnout, which was subsequently associated with increased turnover intention, reduced job satisfaction, and reduced organizational commitment. Similarly, Wing et al. [54] reported that workplace incivility was significantly associated with poor mental health in 394 newly employed nurses in Canada. Incivility from colleagues and supervisors was associated with poor psychological well-being. Lim et al. [55] reported that repeated uncivil behaviors were associated with a toxic work environment and increased depressive and anxiety symptoms that may persist for several years. Qiu and Zhang [56] indicated that workplace incivility, perceived servant leadership, and their interaction were associated with psychological distress among nurses. Overall, these findings suggest consistent associations between workplace incivility and adverse psychological and occupational outcomes across different nursing populations. Nurses reporting higher workplace incivility had lower job control, lower workplace justice, higher psychological job demands, greater burnout, poorer mental health, and poorer self-rated health. Therefore, higher workplace incivility is associated with a less favorable psychosocial work environment and poorer health outcomes in nurses.

In the present study, low or moderate job control and low workplace justice were significantly associated with high workplace incivility among nurses. Nursing is a demanding and high-risk profession that requires

close collaboration among health-care professionals; therefore, a respectful and supportive work environment is essential [57]. According to the Job Demands–Resources model, job control and workplace justice are crucial job resources that may help employees manage work demands, maintain motivation, and cope with occupational stress [30]. Junaković and Macuka [58] reported that job control was the only significant predictor of occupational well-being. Greater job control was associated with higher job satisfaction and lower disengagement. Navajas-Romero et al. [59] reported that job control moderated the association between psychological job demands and occupational stress. Greater decision-making autonomy may therefore be associated with better mood, vitality, and overall well-being and with lower levels of stress and overwork [59]. By contrast, lower perceived organizational justice is associated with greater moral distress and burnout [34, 60] and higher turnover intention, more counterproductive work behaviors, and poorer mental health [61] in nurses. Consistent with these findings, the present study revealed that job control and workplace justice, rather than demographic characteristics, were significantly associated with workplace incivility. The findings highlight the importance of organizational factors in nurses' experiences of workplace incivility.

Although organizational resources were closely associated with workplace incivility, psychological job demands were also independently associated with adverse health outcomes in nurses. Excessive psychological job demands, such as unpredictable tasks, heavy workloads, and frequent interpersonal conflicts, have been consistently associated with poor self-rated health, high burnout, and reduced mental well-being [35]. In nurses, these associations may be intensified by complex role expectations, high responsibility, and limited decision-making authority [62]. Excessive psychological workload is associated with workplace incivility and may be accompanied by poor emotional regulation, low interpersonal tolerance, and high irritability [63]. Similarly, emotionally demanding work environments and heavy workloads have been associated with frequent workplace mistreatment and interpersonal tension [64, 65]. Consistent with previous studies, the present study indicated that workplace incivility and psychological job demands were independently associated with high burnout, poor mental health, and poor self-rated health. In this study, nurses working rotating or night shifts reported lower burnout than those working day shifts. However, previous findings on the association between shift work and burnout have been inconsistent. For example, Gu et al. [66] reported elevated burnout and cynicism in nurses working irregular shifts. Dall'Pra et al. [67] identified long work hours, inadequate staffing, and limited schedule control as factors associated with burnout, although these associations were not retained in multivariable analyses. By contrast, Chiang et al. [68] found no significant association between shift work and burnout after adjustment for work-related factors. Qtait et al. [69] identified sleep quality as a crucial mediator of the association between rotating shift work and burnout, suggesting that sleep partly explains this association. Future longitudinal studies should examine the role of sleep quality in the association between shift work and burnout among nurses. In the present study, job seniority was associated only with poor mental health, suggesting that less experienced nurses are particularly vulnerable during their transition to clinical practice. Newly graduated nurses often experience difficulties with time management, heavy workloads, and interpersonal challenges as they adapt to their professional roles [70]. Similarly, Shi et al. [71] reported high levels of anxiety and emotional distress among new nurses in China, which may reflect their limited familiarity with clinical workflows and technical skills. Overall, these findings indicate that both psychological job demands and workplace incivility were independently associated with adverse health outcomes among nurses. The findings also highlight the need to address both interpersonal and organizational psychosocial factors to support nurses' well-being and care quality. Health-care organizations should consider implementing zero-tolerance policies toward workplace incivility, providing assertive communication and cognitive rehearsal training, improving workload distribution to reduce psychological job demands, and promoting workplace justice and job autonomy. These strategies may help create a respectful and psychologically safe work environment and support nurse well-being, retention, and the delivery of high-quality patient care.

This study has several limitations. First, the cross-sectional design precludes causal inference regarding the relationships between workplace incivility, psychosocial work characteristics, and health outcomes. Although significant associations were identified, the temporal sequence and direction of these associations cannot be

determined. Therefore, longitudinal or prospective studies are required to clarify temporal relationships and potential causal pathways. Second, this study included only currently employed nurses from a single medical center in northern Taiwan, which may have introduced survivor bias and limited the generalizability of the findings. Nurses who had left the profession because of severe workplace incivility, burnout, or poor health were not represented. Therefore, the observed associations may have been underestimated. Furthermore, organizational culture, leadership practices, staffing patterns, workplace policies, and work environments vary across health-care institutions and geographic regions. These differences may affect both exposure to workplace incivility and health outcomes. Therefore, the findings should be interpreted with caution when applied to nurses in other organizational or regional settings. Additional multicenter studies involving diverse health-care institutions and geographic regions are required to improve the generalizability of the findings. Third, all variables were measured using self-report questionnaires, which may have introduced recall bias, common method bias, and social desirability bias. The modest internal consistency of the job control measure may also have attenuated its associations with the health outcomes. In addition, the Workplace Incivility Scale requires participants to recall uncivil experiences over the preceding 12 months. This extended recall period may be subject to memory distortion and selective recall, which could affect prevalence estimates and contribute to overestimation of workplace incivility [3, 4]. The difference in reference periods may also have affected the observed associations: workplace incivility was assessed over the preceding 12 months, whereas burnout and mental health were assessed over the preceding week. Future studies should consider shorter recall periods, experience-sampling methods, or prospective longitudinal designs. They could also incorporate objective indicators, such as sickness absence, turnover records, physiological stress biomarkers (e.g., cortisol), supervisor-rated outcomes, and measures of sleep quality, to improve measurement accuracy and reduce reporting bias. Fourth, although this study simultaneously controlled for psychological job demands, job control, and workplace justice, several potentially important organizational and individual factors were not assessed—for example, staffing adequacy, coworker and supervisor support, leadership style, organizational support, resilience, coping strategies, and previous mental health conditions. These factors may confound or modify the associations between workplace incivility and health outcomes in nurses. Future studies should consider these factors to provide a more comprehensive understanding of the relationship between workplace incivility and nurse well-being. Finally, workplace incivility and health outcomes were analyzed as dichotomous variables. Although this approach reflected the clinically defined nature of the outcomes and used validated cutoff scores, dichotomization may result in information loss and reduced statistical power. However, a robustness analysis using continuous scores revealed associations that were consistent in direction and statistical significance with those of the main analyses (Additional File 1).

Despite its limitations, this study makes several valuable contributions to the literature. Previous studies have often examined work characteristics separately, whereas the present study included multiple psychosocial work characteristics in a single model. This approach enabled a simultaneous examination of the independent associations of various work characteristics with health outcomes in nurses. The use of validated instruments to measure each construct also strengthened the reliability of the findings. Furthermore, most studies on workplace incivility have been conducted in Western settings. The present study provides evidence from a Taiwanese sample and thus offers a non-Western perspective. Cultural norms related to organizational behavior and interpersonal interactions may affect how workplace incivility is experienced and reported. Therefore, our findings highlight the importance of considering cultural context when examining the associations between workplace psychosocial hazards and nurse health.

Conclusion

This study examined the association of workplace incivility with nurse health. Workplace incivility remained significantly associated with poor mental health, high burnout, and poor self-rated health after adjustment for psychological job demands, job control, and workplace justice. Psychological job demands were also significantly

associated with all three health outcomes. These findings indicate that workplace incivility and high psychological demands are crucial psychosocial factors associated with poor health in nurses. Health-care organizations should devise strategies to promote a respectful and supportive work environment, establish clear policies to address workplace incivility, optimize workload distribution, and support nurses' resilience and overall well-being. In the future, longitudinal studies should be conducted to establish causal relationships and evaluate the effectiveness of various organizational strategies.

Abbreviations

CI Confidence interval
OR Odds ratio

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate This study was conducted in accordance with the ethical principles of the Declaration of Helsinki. The study protocol was approved by the Joint Institutional Review Board of Taipei Medical University (approval number: N202209026). The requirement for informed consent was waived by the Joint Institutional Review Board of Taipei Medical University because the survey was anonymous and voluntary and posed minimal risk to participants.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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